Dec. 2

Due Thurs.

HW #14

HW #15 ← good effort only

Due Friday

Take-home Test 3

projects (MATH 540 only)

Wave eq'n in bounded domain for $n=2$ (Not responsible for)

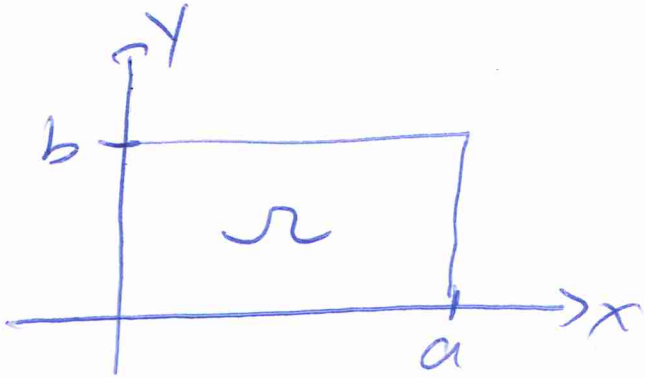
<u>2-D</u>

$$\Delta u - u_{tt} = 0 \quad \underline{x} \in \Omega, \; t > 0$$

$$u(\underline{x},0) = \phi(\underline{x}) \quad \underline{x} \in \Omega$$

$$u_t(\underline{x},0) = \psi(\underline{x})$$

$$u = 0 \quad \underline{x} \in \partial\Omega \quad t > 0$$



can solve using of variables

$$u_{mn} = \underbrace{\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)}_{\text{eigenfunctions}}\left[A_{mn}\cos\sqrt{\lambda_{mn}}\,t + B_{mn}\sin\left(\sqrt{\lambda_{mn}}\,t\right)\right]$$

$$u(x,t) = \sum_{m,n=1}^{\infty} u_{mn}$$

eigenvalue $\rightarrow \lambda_{mn} = \pi^2\left[\frac{m^2}{a^2} + \frac{n^2}{b^2}\right]$

constants:

$$\phi(x,y) = \sum_{m,n=1}^{\infty} A_{mn}\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)$$

$$\psi(x,y) = \sum_{m,n=1}^{\infty} \sqrt{\lambda_{mn}}\, B_{mn} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right)$$

<u>Remark</u>

in Lin Alg $\quad A\underline{x} = \lambda \underline{x} \quad \underline{x} \neq \underline{0}, \quad \lambda$ = eigenvalue

$\underline{x}$ = eigenvector

Linear operator L

$LV = \lambda V \qquad V$, is a ~~fun.~~ f'n

λ = eigenvalue $\qquad V$ is an eigen f'n

$LV + \lambda V = 0 \qquad$ (sign is not strict for λ)

Review + Heat Eq'n

① Laplace eq'n $\Delta u = 0$

Physical problem for which Laplace eq'n is a good mathematical model

steady-state (no change in time) of temp. distrib.

② Properties of sol'n (harmonic f'ns)

① $u \in C^0(\Omega) \Rightarrow u \in C^\infty(\Omega)$

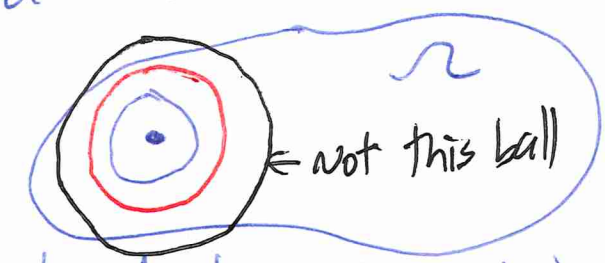
② Maximum Principle

$u \in C^0(\bar{\Omega})$ + u is harmonic, + Ω is bdd

$\Rightarrow$ u attains its max + min values on $\partial\Omega$

+ only on $\partial\Omega$ unless u = constant.

③ Mean value property.



$u(\underline{x})$ is the avg of the temp of any sphere centered at $\underline{x}$ + contained in Ω.

What problems are well-posed?

Find $u \in C^0(\bar{\Omega})$ (Ω is bdd) such that

$$\Delta u = 0 \text{ on } \Omega$$

and

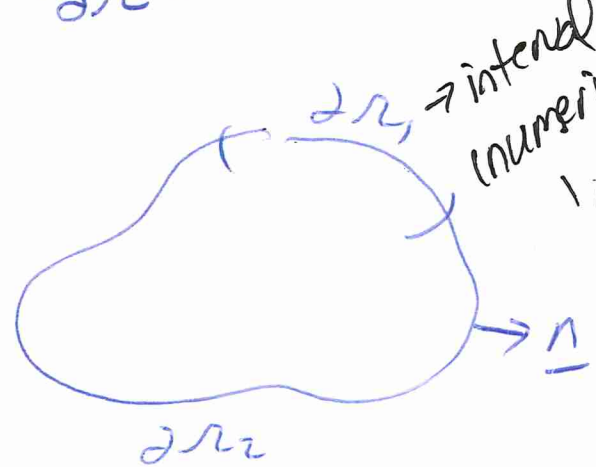
① $u(\underline{x}) = f(\underline{x})$ for $\underline{x} \in \partial\Omega$, $f(\underline{x}) \in C^0(\partial\Omega)$

② $\frac{\partial u}{\partial \underline{n}}(\underline{x}) = g(\underline{x})$ for $\underline{x} \in \partial\Omega$, $g(\underline{x}) \in C^0(\partial\Omega)$

$$\int_{\partial\Omega} g(\underline{x})\, d\underline{x} = 0$$

uniqueness only up to a constant

③ $\partial\Omega = \partial\Omega_1 \cup \partial\Omega_2$



$u(\underline{x}) = f(\underline{x})$ $\underline{x} \in \partial\Omega_1$

$\frac{\partial u}{\partial n}(\underline{x}) = g(\underline{x})$ $\underline{x} \in \partial\Omega_2$

$f(\underline{x}) \in C^0(\partial\Omega_1)$ $g(\underline{x}) \in C^0(\partial\Omega_2)$

well-posed

No integral condition on $g(\underline{x})$ is needed.

Heat Eq'n

1-Dimension

$$u_t - \Delta u = 0$$

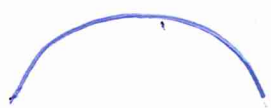
Physical

pipe is insulated on sides only

+ Thin temp ≈ constant over ~~Are~~ all cross-section

example

$$u(0,t) = u(L,t) = 0$$

$u(x,0) =$  quadratic

Well-posed problem

IBV problems (Initial, Boundary Value)

Find $u(x,t)$ continuous in the rectangle

$$0 \le x \le L, \quad 0 \le t < \infty$$

$$u_t - \Delta u = 0 \quad 0 < x < L, \; 0 < t$$

$$u(x,0) = \phi(x) \quad 0 \le x \le L$$

$$\left.\begin{aligned} u(0,t) &= f_1(t) \\ u(L,t) &= f_2(t) \end{aligned}\right\} \; t > 0$$

where ϕ, f_1, f_2 are continuous f'ns.
that satisfy the compatibility condition

$$\phi(0) = f_1(0) \qquad \phi(L) = f_2(0)$$

How to solve?

separation of variables

$\Rightarrow$ $\underbrace{\text{Fourier series}}_{\text{disk}}$ or $\underbrace{\text{some version of it}}_{\text{rectangle}}$